

DR MAYUKH PAHARI

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For research and other activity updates, check at <https://mayukhpahari.wixsite.com/website>

1.1 Employment

2020-present	Assistant Professor, Department of Physics, Indian Institute of Technology, Hyderabad, Kandi, Telangana 502285
2018-2020	Newton International Fellow, School of Physics and Astronomy, University of Southampton: <i>X-ray reverberation and X-ray/UV/optical reverberation mapping in Seyfert galaxies</i>
2015-2018	Postdoctoral Researcher, Inter-University Centre for Astronomy and Astrophysics (IUCAA), India: <i>AstroSat/LAXPC calibration, pipeline testing and spectra-timing analysis of X-ray binaries as observed by AstroSat</i>
2011 - 2015	Senior Research Fellow, Department of Astronomy and Astrophysics, Tata Institute of Fundamental Research, Mumbai, India, : <i>spectro-timing behaviour of black hole X-ray binaries during X-ray state transitions</i>
2008 - 2011	Junior Research Fellow, Department of Astronomy and Astrophysics, Tata Institute of Fundamental Research, Mumbai, India, : <i>Characterising the room-temperature semiconductor detectors suitable for space mission and exploring properties of X-ray binaries</i>

1.2 Education

2009-2015	PhD in Astrophysics, <i>Understanding accretion and radiation process leading to state transition in X-ray binaries</i> , Research center: Department of Physics, Tata Institute of Fundamental Research, Homi Bhabha Road, Colaba, Mumbai 400005 Thesis Submission center: School of Studies in Physics and Astrophysics, TIFR, Mumbai and Pt. Ravishankar Shukla University, Raipur, India
2006-2008	Master's in Physics and Astrophysics, Department of Physics, University of Pune and IUCAA; Master's thesis title: <i>X-ray data analysis of compact Galactic objects</i>
2003-2006	Bachelor of Science, University of Calcutta, Physics (Hons.)

Total Refereed publications: 62 [**19 first author & 43 coauthor**]

Journal (impact factor): Nature (49.8), MNRAS (5.23), MNRAS Letters (5.42),
ApJ (5.58), ApJ Letters (8.37), Journal of high energy astrophysics (10.5),
Nature Astronomy (14.5), JoAA (1.27), NIMPA (1.26)

1.3 Publications in the last 5 years (2020-2025)

[1] Mondal, A. S., Pahari, M., and Dewangan, G. C., "NuSTAR discovers a long type-I X-ray burst from the clocked burster GS 1826-24", Journal of High Energy Astrophysics, vol. 49, Art. no. 100450, Elsevier, 2026. doi:10.1016/j.jheap.2025.100450.

- [2] Chatterjee, K., “Distinct Fe K Line Complexes in MAXI J1744-294 Revealed by XRISM High-resolution Spectroscopy”, *The Astrophysical Journal*, vol. 992, no. 2, Art. no. 210, IOP, 2025. doi:10.3847/1538-4357/ae07d8
- [3] Bagchi, M., “Astrophysics with compact objects: An Indian perspective, present status and future vision”, *Journal of Astrophysics and Astronomy*, vol. 46, no. 2, Art. no. 62, Springer, 2025. doi:10.1007/s12036-025-10077-5
- [4] Dewangan, G. C., “Supermassive black holes and active galactic nuclei”, *Journal of Astrophysics and Astronomy*, vol. 46, no. 2, Art. no. 60, Springer, 2025. doi:10.1007/s12036-025-10075-7
- [1] Baruah, A. and Pahari, M., “UV Flux Variation Study in Contact Binary VW Cephei” 2025 *Res. Astron. Astrophys.* 25 105013 DOI 10.1088/1674-4527/adf9da
- [5] Sethuram, R., “New age observing facilities for Indian astronomy: 2020–2035”, *Journal of Astrophysics and Astronomy*, vol. 46, no. 1, Art. no. 25, Springer, 2025. doi:10.1007/s12036-025-10047-x.
- [6] Mondal, A. S., Bhattacharya, M., Pahari, M., Raychaudhuri, B., Ghosh, R., and Dewangan, G. C., “NuSTAR view of the X-ray transients Swift J174805.3-244637 and IGR J17511-3057”, *Journal of High Energy Astrophysics*, vol. 45, Elsevier, pp. 359–370, 2025. doi:10.1016/j.jheap.2025.01.005.
- [7] Sarkar, A. and Pahari, M., “Effect of magnetic field on the formation of radiatively inefficient accretion flow around black holes”, *Physical Review D*, vol. 111, no. 2, Art. no. 023008, APS, 2025. doi:10.1103/PhysRevD.111.023008.
- [8] Dutta, T., Pahari, M., Sarkar, A., Bhattacharyya, S., and Bhargava, Y., “Probing the origin of the extended flaring branch of Z-type X-ray binaries GX 340+0 and GX 5-1 using AstroSat”, *Monthly Notices of the Royal Astronomical Society*, vol. 535, no. 4, OUP, pp. 3383–3395, 2024. doi:10.1093/mnras/stae2529.
- [9] Bhattacharya, M., Mondal, A. S., Pahari, M., Raychaudhuri, B., Ghosh, R., and Dewangan, G. C., “Relativistic X-ray reflection from the accreting millisecond X-ray pulsar IGR J17498-2921”, *Monthly Notices of the Royal Astronomical Society*, vol. 534, no. 1, OUP, pp. 634–644, 2024. doi:10.1093/mnras/stae2135.
- [10] Pahari, M., “AstroSat and NICER timing view of the Z-type neutron star X-ray binary GX 340 + 0”, *Monthly Notices of the Royal Astronomical Society*, vol. 528, no. 3, OUP, pp. 4125–4138, 2024. doi:10.1093/mnras/stae309.
- [11] Bhargava, Y., Bhattacharyya, S., Homan, J., and Pahari, M., “AstroSat View of the Neutron Star Low-mass X-Ray Binary GX 340+0”, *The Astrophysical Journal*, vol. 955, no. 2, Art. no. 102, IOP, 2023. doi:10.3847/1538-4357/acee7a.
- [12] Vincentelli, F. M., Beard, M., Hardy, I. M., Cackett, E., Horne, K., and Pahari, M., “The evolution of the UV/optical lag spectrum of NGC 7469 seen by the Liverpool Telescope”, *Astronomische Nachrichten*, vol. 344, no. 4, Art. no. e20230018, 2023. doi:10.1002/asna.20230018.
- [13] Beard, M. W. J., “Time-scale-dependent X-ray to UV time lags of NGC 4593 using high-intensity XMM-Newton observations with Swift and AstroSat”, *Monthly Notices of the Royal Astronomical Society*, vol. 519, no. 1, OUP, pp. 91–101, 2023. doi:10.1093/mnras/stac3391.
- [14] **Castro Segura, N...Pahari, M. et al., “A persistent ultraviolet outflow from an accreting neutron star binary transient”, *Nature*, vol. 603, no. 7899, pp. 52–57, 2022. doi:10.1038/s41586-021-04324-2.**
- [15] Williams, D. R. A., “LeMMINGs - IV. The X-ray properties of a statistically complete sample of the nuclei in active and inactive galaxies from the Palomar sample”, *Monthly Notices of the Royal Astronomical Society*, vol. 510, no. 4, OUP, pp. 4909–4928, 2022. doi:10.1093/mnras/stab3310.
- [16] Baldi, R. D., “LeMMINGs III. The e-MERLIN legacy survey of the Palomar sample: exploring the origin of nuclear radio emission in active and inactive galaxies through the [O III] - radio connection”, *Monthly Notices of the Royal Astronomical Society*, vol. 508, no. 2, OUP, pp. 2019–2038, 2021. doi:10.1093/mnras/stab2613.
- [17] Yadav, J. S., Agrawal, P. C., Misra, R., Roy, J., Pahari, M., and Manchanda, R. K., “LAXPC instrument onboard AstroSat: Five exciting years of new scientific results specially on X-ray binaries”, *Journal of Astrophysics and Astronomy*, vol. 42, no. 2, Art. no. 40, Springer, 2021. doi:10.1007/s12036-021-09717-3.
- [18] Antia, H. M., “Large Area X-ray Proportional Counter (LAXPC) in orbit performance: Calibration, background, analysis software”, *Journal of Astrophysics and Astronomy*, vol. 42, no. 2, Art. no. 32, Springer, 2021. doi:10.1007/s12036-021-09712-8.

- [19] Vincentelli, F. M., “On the multiwavelength variability of Mrk 110: two components acting at different time-scales”, *Monthly Notices of the Royal Astronomical Society*, vol. 504, no. 3, OUP, pp. 4337–4353, 2021. doi:10.1093/mnras/stab1033.
- [20] Castro Segura, N., “Bow shocks, nova shells, disc winds and tilted discs: the nova-like V341 Ara has it all”, *Monthly Notices of the Royal Astronomical Society*, vol. 501, no. 2, OUP, pp. 1951–1969, 2021. doi:10.1093/mnras/staa2516.
- [21] Williams, D. R. A., “The curious activity in the nucleus of NGC 4151: jet interaction causing variability?”, *Monthly Notices of the Royal Astronomical Society*, vol. 495, no. 3, OUP, pp. 3079–3086, 2020. doi:10.1093/mnras/staa1152.
- [22] Pahari, M., “Evidence for variability time-scale-dependent UV/X-ray delay in Seyfert 1 AGN NGC 7469”, *Monthly Notices of the Royal Astronomical Society*, vol. 494, no. 3, OUP, pp. 4057–4068, 2020. doi:10.1093/mnras/staa1055.
- [23] Vincentelli, F. M., Mastroserio, G., McHardy, I., Ingram, A., and Pahari, M., “X-ray reverberation lags from the 1.5 Seyfert galaxy NGC 5273”, *Monthly Notices of the Royal Astronomical Society*, vol. 492, no. 1, OUP, pp. 1135–1141, 2020. doi:10.1093/mnras/stz3511.

Link to Full List of Publications: [https://ui.adsabs.harvard.edu/search/fq=%7B!type%3Daqp%20v%3D%24fq_database%7D&fq_database=\(database%3Aastronomy%20OR%20database%3Aphysics\)&q=%20author%3A%22Pahari%2C%20M.%22&sort=date%20desc%2C%20bibcode%20desc&p_0](https://ui.adsabs.harvard.edu/search/fq=%7B!type%3Daqp%20v%3D%24fq_database%7D&fq_database=(database%3Aastronomy%20OR%20database%3Aphysics)&q=%20author%3A%22Pahari%2C%20M.%22&sort=date%20desc%2C%20bibcode%20desc&p_0)

1.4 Awarded fellowships

2018 - 2020	Newton International Fellowship by the Royal Society, UK, hosted by Prof Ian McHardy, University of Southampton
2011 - 2014	Senior Research Fellowship , Department of Astronomy and Astrophysics, Tata Institute of Fundamental Research, Mumbai, India, supervised by: Prof J S Yadav
2008 - 2011	Junior Research Fellowship , Department of Astronomy and Astrophysics, Tata Institute of Fundamental Research, Mumbai, India, Supervised by: Prof J S Yadav

1.5 Funded project as Principal Investigator

2021 - 2024	Establishing Advanced Darksky Observatory at IIT Hyderabad (seed Research funding: 34 lakhs)
2025 - 2027	Numerical simulation and experimental validation of Low Earth Object (Funded by: DRDO, India: 180 lakhs)

1.4 Organisational and professional activities

2025-present	Awarded Honorary/associate faculty position at the Indian Centre for Space Physics (ICSP), Kolkata
2024-present	Joined as the guest editor of the Springer Nature journal Discover Space
2023	Awarded life membership of Astronomical Society of India
2022	Established Advanced Darksky Observatory , first oncampus 0.5m observatory among all IITs

2019	Press release : Southampton University Press ' <i>University researchers discover black hole in our Galaxy spinning rapidly around itself</i> ' Details: https://www.southampton.ac.uk/news/2019/01/black-hole-spinning.page
2017-present	International Science Development Team member, Thirty Meter Telescope (TMT) project, TMT corporation, California, USA
2016-present	Working as the reviewer in the Monthly Notice of Royal Astronomical Society and Advances in Space Research
2016-present	Working as a proposal reviewer for AstroSat mission
2015-present	Instrument team member of LAXPC/AstroSat, India's first multi wavelength space-based Astronomical satellite, India
2013 - 2014	LAXPC instrument testing and calibration assistant, TIFR, India
2008 - 2011	X-ray instrument lab assistant at the Astronomy group, TIFR, India

1.5 Major current collaborations

2018 - present	International Chandra and e-MERLIN legacy survey of nearby galaxies (LeM MINGs survey) [University of Southampton, Oxford and Liverpool]
2018 - present	Simultaneous Multiwavelength Astronomy Research in Transients Network (SmartNet) [ESO and Canada collaborative project]
2017 - 2018	ESO astronomical research in Southern Hemisphere (revisiting black holes)

1.6 Data analysis skills

RXTE, AstroSat, IUE, Swift (XRT/UVOT), XMM-Newton, Chandra, NuSTAR, NICER

1.7 Programming and Numerical Skills

Language skills: Fortran, Python and IDL

- Developed Python code for X-ray lag and frequency-resolved RMS spectral analysis
- Developed Python code for X-ray lag spectral model fitting
- Developed and tested AstroSat/LAXPC data pipeline software

1.8 Observing experience

- **Principal Investigator of 17 approved programmes with 359 hours of international observation** in AstroSat, Swift, GMRT, INTEGRAL, XMM-Newton, e-Merlin and SALT
- Co-PI of 23 approved programmes with 513 hours in AstroSat, HST, SALT, Swift, Ultracam

1.9 Teaching and mentoring experience

Courses Taught in the last five academic years (2020-2024)

(A) Nuclear Physics (B) Quantum Mechanics (C) Electronic circuit theory (D) Atomic and Molecular physics (E) X-ray physics (F) Concepts and hands-on Astrophysics (G) Modern Physics

Current PhD students

Name	Year of Joining	Funding type	Thesis title	Current Status
Ms Shree Suman	2020	CSIR-JRF	Understanding coronal geometry through variability studies	Thesis writing: OC proposed in December 2025
Mr Saurabh Sharma	2022	MoE-funded	Chandra non-nuclear survey of nearby galaxy field	Ongoing
Mr Vishal Jadoliya	2023	MoE-funded	Modelling X-ray polarisation geometry in X-ray binaries	Ongoing
Mr Anurag Baruah	2024	DST-INSPIRE fellow	Modelling mass transfer of contact binary stars	Ongoing
Ms Megha Tomer	2024	UGC-JRF	Planetary disc formation modelling and measurements	Ongoing
Mr Sourav Paul	2025	External Project Funded	LEO orbit determination with AI/ML assisted techniques	Ongoing

Other mentoring experiences (within IIT Hyderabad)

Top 10 Master's Thesis supervision (Out of 15)

Student's Name	Year	Roll No	Thesis Title	Student's current status
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Priyansh Jaswal	2021	PH20MSCST11006	Understanding the nature of complex X-ray variabilities in the Galactic Microquasar GRS 1915+105	Research scholar in IISER, Kolkata
Shuvajit Khatua	2022	PH21MSCST11007	Understanding the nature of X-ray spectra from black hole systems	Research scholar in RRI, Bangalore
Tanmoy Dutta	2023	PH22MSCST11022	On the origin of the extended flaring branch in Z-type X-ray binaries GX 340+0 and GX 5-1 using AstroSat	Research scholar in IISC, Bangalore
Anish Sarkar	2023	PH22MSCST11032	Structure of Magnetic fields in launching jets in SMBH	Research scholar in ANU, Australia
Sumanta Biswas	2023	PH22MSCST11005	Attempting phase-resolved spectroscopy for a Quasi Periodic Oscillation (QPO)	Research scholar in GTIIT, China
Ravina Paldiya	2023	PH22MSCST11034	UV spectral analysis of Active Galactic Nuclei	Lecturer at Sri Chaitanya college, Hyderabad
Neelanjana Dey	2024	PH23MSCST11023	Analysing variable star using ML based feature detection in X-ray binaries	Research Scholar in IIT Madras
K M Tanu	2024	PH23MSCST11020	Exploring state transitions in black hole X-ray binaries with IXPE	Research analyst at Robust Results Put Ltd., an MoE-IITK collaboration

Vivek Shinde	2024	PH23MSCST11040	AGN variability study in multi wavelength using Damped Random Walk model	Lecturer at GreenValley Junior College, Warangal
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Btech Project supervision (within IITH): 15

Other mentoring and lecturing experiences (outside IIT Hyderabad)

2024-2025	Supervised MSc Thesis of Ms Athira Mohanan, Central University of Karnataka
2023-2024	Supervised BS-MS thesis of Mr Harikumar N, NIT Rourkela
2022-2023	Supervised BS-MS thesis of Mr Romeo Pallikkara, IISER Mohali
2019	Delivered a lecture series on X-ray instrumentation and conducted hands-on tutorial sessions on AstroSat data analysis at the University of Southampton, UK
2019	co-mentored Mr. Alex Weston, the Royal Society-funded summer undergraduate, University of Southampton
2013 - 2014	Mentored Mr. Jai Verdhhan Chauhan, postgraduate at TIFR, Mumbai, India
2016 - 2017	co-mentored Ms D Rawat, PhD, Indian Institute of Technology, Kanpur
2016 - 2018	co-mentored Dr Aditya Sow Mondal, PhD, Viswa Bharati University, India

2.0 Few selected Invited and solicited talks

2025	Invited talk at the High energy consortium, Suranaree University of Technology, Thailand
2023	Invited talk at the conference on 'Recent Development in X-ray astronomy' at 21-25 October, 2023, Fudan University, Shanghai, China
2022	Invited talk at the 23rd COSPAR International Assembly, Athens, Greece
2022	Invited talk at the conference on AstroSat's 5 years, 14-16 February ISRO headquarters, Bangalore
2019	Solicited talk at the 'Future of X-ray Timing' conference, Amsterdam, the Netherlands
2019	Solicited talk at the 'Mapping central regions of Active Galactic Nuclei' conference, Guilin, China
2019	Solicited talks at the SPINS-UK meeting at the University College, London
2018	Solicited talks at the TMT forum, 'Extremely big eyes on the early universe', University of California, Los Angeles

- 2017 Solicited talk at the conference on “Power of X-ray spectroscopy” at the Warsaw, Poland
- 2016 Invited participant at ‘The X-ray Spectral-Timing Revolution’ at the Lorentz Centre, University of Leiden, Netherlands

Refereed Publication [60]**[22 as a first/corresponding author and 38 as co-author]**

Journals in which papers are published:

MNRAS - Monthly Notices of Royal Astronomical Society, Oxford University Press, UK, Impact factor: 5.231 [7 papers published as a lead author]**MNRAS Letters**, Oxford University Press, impact factor: 5.42 [1 paper published as a lead author]**ApJ** - Astrophysical Journal, IoP Publishing, USA, impact factor: 5.58 [5 papers published as a lead author]**ApJ Letters**, IoP Publishing, USA, impact factor: 8.37 [1 paper published as a lead author]**Nature**: impact factor 48.5**Journal of High Energy Astrophysics**; impact factor 10.5**Nature Astronomy LETTER**: Nature publishing group, UK; impact factor: 14.5 [1 paper published as a co-author]**Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment**: Elsevier, impact factor: 1.26 [1 paper published as a lead author]

- Mondal, A. S., Pahari, M., and Dewangan, G. C., “NuSTAR discovers a long type-I X-ray burst from the clocked burster GS 1826-24”, Journal of High Energy Astrophysics, vol. 49, Art. no. 100450, Elsevier, 2026. doi:10.1016/j.jheap.2025.100450.
- Baruah, A. and Pahari, M., “UV Flux Variation Study in Contact Binary VW Cephei”, Research in Astronomy and Astrophysics, vol. 25, no. 10, Art. no. 105013, IOP, 2025. doi:10.1088/1674-4527/adf9da.
- Chatterjee, K., “Distinct Fe K Line Complexes in MAXI J1744-294 Revealed by XRISM High-resolution Spectroscopy”, The Astrophysical Journal, vol. 992, no. 2, Art. no. 210, IOP, 2025. doi:10.3847/1538-4357/ae07d8.
- Bagchi, M., “Astrophysics with compact objects: An Indian perspective, present status and future vision”, Journal of Astrophysics and Astronomy, vol. 46, no. 2, Art. no. 62, Springer, 2025. doi:10.1007/s12036-025-10077-5.
- Dewangan, G. C., “Supermassive black holes and active galactic nuclei”, Journal of Astrophysics and Astronomy, vol. 46, no. 2, Art. no. 60, Springer, 2025. doi:10.1007/s12036-025-10075-7.
- Sethuram, R., “New age observing facilities for Indian astronomy: 2020–2035”, Journal of Astrophysics and Astronomy, vol. 46, no. 1, Art. no. 25, Springer, 2025. doi:10.1007/s12036-025-10047-x.
- Mondal, A. S., Bhattacharya, M., Pahari, M., Raychaudhuri, B., Ghosh, R., and Dewangan, G. C., “NuSTAR view of the X-ray transients Swift J174805.3-244637 and IGR J17511-3057”, Journal of High Energy Astrophysics, vol. 45, Elsevier, pp. 359–370, 2025. doi:10.1016/j.jheap.2025.01.005.

- Sarkar, A. and Pahari, M., “Effect of magnetic field on the formation of radiatively inefficient accretion flow around black holes”, *Physical Review D*, vol. 111, no. 2, Art. no. 023008, APS, 2025. doi:10.1103/PhysRevD.111.023008.
- Dutta, T., Pahari, M., Sarkar, A., Bhattacharyya, S., and Bhargava, Y., “Probing the origin of the extended flaring branch of Z-type X-ray binaries GX 340+0 and GX 5-1 using AstroSat”, *Monthly Notices of the Royal Astronomical Society*, vol. 535, no. 4, OUP, pp. 3383–3395, 2024. doi:10.1093/mnras/stae2529.
- Bhattacharyya, M., Mondal, A. S., Pahari, M., Raychaudhuri, B., Ghosh, R., and Dewangan, G. C., “Relativistic X-ray reflection from the accreting millisecond X-ray pulsar IGR J17498-2921”, *Monthly Notices of the Royal Astronomical Society*, vol. 534, no. 1, OUP, pp. 634–644, 2024. doi:10.1093/mnras/stae2135.
- Pahari, M., “AstroSat and NICER timing view of the Z-type neutron star X-ray binary GX 340 + 0”, *Monthly Notices of the Royal Astronomical Society*, vol. 528, no. 3, OUP, pp. 4125–4138, 2024. doi:10.1093/mnras/stae309.
- Bhargava, Y., Bhattacharyya, S., Homan, J., and Pahari, M., “AstroSat View of the Neutron Star Low-mass X-Ray Binary GX 340+0”, *The Astrophysical Journal*, vol. 955, no. 2, Art. no. 102, IOP, 2023. doi:10.3847/1538-4357/acee7a.
- Vincentelli, F. M., Beard, M., Hardy, I. M., Cackett, E., Horne, K., and Pahari, M., “The evolution of the UV /optical lag spectrum of NGC 7469 seen by the Liverpool Telescope”, *Astronomische Nachrichten*, vol. 344, no. 4, Art. no. e20230018, Wiley, 2023. doi:10.1002/asna.20230018.
- Beard, M. W. J., “Time-scale-dependent X-ray to UV time lags of NGC 4593 using high-intensity XMM-Newton observations with Swift and AstroSat”, *Monthly Notices of the Royal Astronomical Society*, vol. 519, no. 1, OUP, pp. 91–101, 2023. doi:10.1093/mnras/stac3391.
- Castro Segura, N., “A persistent ultraviolet outflow from an accreting neutron star binary transient”, *Nature*, vol. 603, no. 7899, pp. 52–57, 2022. doi:10.1038/s41586-021-04324-2.
- Williams, D. R. A., “LeMMINGs - IV. The X-ray properties of a statistically complete sample of the nuclei in active and inactive galaxies from the Palomar sample”, *Monthly Notices of the Royal Astronomical Society*, vol. 510, no. 4, OUP, pp. 4909–4928, 2022. doi:10.1093/mnras/stab3310.
- Baldi, R. D., “LeMMINGs III. The e-MERLIN legacy survey of the Palomar sample: exploring the origin of nuclear radio emission in active and inactive galaxies through the [O III] - radio connection”, *Monthly Notices of the Royal Astronomical Society*, vol. 508, no. 2, OUP, pp. 2019–2038, 2021. doi:10.1093/mnras/stab2613.
- Yadav, J. S., Agrawal, P. C., Misra, R., Roy, J., Pahari, M., and Manchanda, R. K., “LAXPC instrument onboard AstroSat: Five exciting years of new scientific results specially on X-ray binaries”, *Journal of Astrophysics and Astronomy*, vol. 42, no. 2, Art. no. 40, Springer, 2021. doi:10.1007/s12036-021-09717-3.

- Antia, H. M., “Large Area X-ray Proportional Counter (LAXPC) in orbit performance: Calibration, background, analysis software”, *Journal of Astrophysics and Astronomy*, vol. 42, no. 2, Art. no. 32, Springer, 2021. doi:10.1007/s12036-021-09712-8.
- Vincentelli, F. M., “On the multiwavelength variability of Mrk 110: two components acting at different time-scales”, *Monthly Notices of the Royal Astronomical Society*, vol. 504, no. 3, OUP, pp. 4337–4353, 2021. doi:10.1093/mnras/stab1033.
- N. Castro Segura, C. Knigge, J. A. Acosta-Pulido, D. Altamirano, S. del Palacio, J.V. Hernandez Santisteban, **Pahari, M.**, P. Rodriguez-Gil, C. Belardi, D.A.H. Buckley, M.R. Burleigh, M. Childress, R.P. Fender, D.M. Hewitt, D.J. James, R.B. Kuhn, N.P.M. Kuin, J. Pepper, A.A. Ponomareva, M.L. Pretorius, J.E. Rodríguez, K.G. Stassun, D.R.A. Williams, P.A. Woudt 2020, *MNRAS*, Link: <https://arxiv.org/pdf/2008.07462.pdf> : *Bow-shocks, nova shells, disc winds and tilted discs: The Nova-Like V341 Ara Has It All*
- **Pahari, M.**, McHardy, I. M., Vincentelli, F., Cackett, E., Peterson, B. M., Goad, M., Gultekin, K., Horne, K. 2020, *MNRAS*, 494, 4057: Link: <https://ui.adsabs.harvard.edu/abs/2020MNRAS.494.4057P> : *Evidence for variability time-scale dependent UV/X-ray delay in Seyfert 1 AGN NGC 7469*
- Vincentelli, F. M., Mastroserio, G., McHardy, I. M., Ingram, A., **Pahari, M.**, 2020, *MNRAS*, 492, 1135: Link: <https://ui.adsabs.harvard.edu/abs/2020MNRAS.492.1135V> : *X-ray reverberation lags from the 1.5 Seyfert galaxy NGC 5273*
- D. R. A. Williams, McHardy, I. M., Baldi, R. D., Beswick, R. J., **Pahari, M.**, Argo, M. K., Beri, A. et al., 2019, *MNRAS*, 486, 4962: Link: <https://ui.adsabs.harvard.edu/abs/2019MNRAS.486.4962W> : *Unveiling the 100 pc scale nuclear radio structure of NGC 6217 with e-MERLIN and the VLA*
- Beri, Aru, Paul, Biswajit, Yadav, J. S., Antia, H. M., Agrawal, P. C., Manchanda, R. K., Dedhia, Dhiraj, Chauhan, Jai Verdhana, **Pahari, M.**, Misra, Ranjeev, et al. 2019, *MNRAS*, 482, 4397: Link: <https://ui.adsabs.harvard.edu/abs/2019MNRAS.482.4397B> : *Thermonuclear X-ray bursts in rapid succession in 4U 1636-536 with AstroSat-LAXPC*
- Roy, J., Agrawal, P. C., Iyer, N. K., Bhattacharya, D., Yadav, J. S., Antia, H. M., Chauhan, J. V., Choudhury, M., Dedhia, D. K., Katoch, T., Madhavani, P., Manchanda, R. K., Misra, R., **Pahari, M.**, Paul, B., and Shah, P. 2019, *ApJ*, 872, 33: Link: <https://ui.adsabs.harvard.edu/abs/2019ApJ...872...33R> : *LAXPC/AstroSat Study of ~1 and ~2 mHz Quasi-periodic Oscillations in the Be/X-Ray Binary 4U 0115+63 during Its 2015 Outburst*
- Rawat, Divya, **Pahari, M.**, Yadav, J. S., Jain, Pankaj, Misra, Ranjeev et al. 2019, *ApJ*, 870, 4: Link: <https://ui.adsabs.harvard.edu/abs/2019ApJ...870....4R> : *Study of Timing Evolution from Nonvariable to Structured Large-amplitude Variability Transition in GRS 1915 + 105 Using AstroSat*
- McHardy, I. M., Connolly, S. D., Horne, K., Cackett, E. M., Gelbord, J., Peterson, B. M., **Pahari, M.**, Gehrels, N., et al., 2018, *MNRAS*, 480, 2881: Link: <https://ui.adsabs.harvard.edu/abs/2018MNRAS.480.2881M> : *X-ray/UV/optical variability of NGC 4593 with Swift: reprocessing of X-rays by an extended reprocessor*
- **Pahari, M.**, Bhattacharyya, S., Rao, A. R., Bhattacharya, D, et al. 2018, *ApJ*, 867, 86 :

Link: <https://ui.adsabs.harvard.edu/abs/2018ApJ...867...86P> : *AstroSat and Chandra View of the High Soft State of 4U 1630-47 (4U 1630-472): Evidence of the Disk Wind and a Rapidly Spinning Black Hole*

- Mondal, Aditya S., Dewangan, G. C., **Pahari, M.**, and Raychaudhuri, B. 2018, MNRAS, 474, 2064: Link: <https://ui.adsabs.harvard.edu/abs/2018MNRAS.474.2064M> : *NuSTAR view of the Z-type neutron star low-mass X-ray binary Cygnus X-2*
- **Pahari, M.**, Yadav, J. S., Verdhhan Chauhan, Jai, Rawat, Divya, et al., 2018, ApJ Letters, 853, 11 : Link: <https://ui.adsabs.harvard.edu/abs/2018ApJ...853L..11P> : *Extensive Broadband X-Ray Monitoring During the Formation of a Giant Radio Jet Base in Cyg X-3 with AstroSat*
- **Pahari, M.**, Antia, H. M., Yadav, J. S., et al. 2017, ApJ, 849, 16: Link: <https://ui.adsabs.harvard.edu/abs/2017ApJ...849...16P> : *X-Ray Timing Analysis of Cyg X-3 Using AstroSat/LAXPC: Detection of Milli-hertz Quasi-periodic Oscillations during the Flaring Hard X-Ray State*
- Gandhi, P., Bachetti, M., Dhillon, V. S., Fender, R. P., Hardy, L. K., Harrison, F. A., Littlefair, S. P., Malzac, J., Markoff, S., Marsh, T. R., Mooley, K., Stern, D., Tomsick, J. A., Walton, D. J., Casella, P., Vincentelli, F., Altamirano, D., Casares, J., Ceccobello, C., Charles, P. A., Ferrigno, C., Hynes, R. I., Knigge, C., Kuulkers, E., **Pahari, M.**, et al. 2017, Nature Astronomy, 1, 859 : Link: <https://ui.adsabs.harvard.edu/abs/2017NatAs...1..859G> : *An elevation of 0.1 light-seconds for the optical jet base in an accreting Galactic black hole system*
- **Pahari, M.**, McHardy, I. M., Mallick, L., Dewangan, G. C., and Misra, R. 2017, MNRAS, 470, 3239: Link: <https://ui.adsabs.harvard.edu/abs/2017MNRAS.470.3239P> : *Detection of the high-energy cut-off from the Seyfert 1.5 galaxy NGC 5273*
- **Pahari, M.**, Gandhi, Poshak, Charles, Philip A., Kotze, Marissa M., Altamirano, D. and Misra, Ranjeev 2017, MNRAS, 469, 193: Link: <https://ui.adsabs.harvard.edu/abs/2017MNRAS.469..193P> : *Simultaneous optical/X-ray study of GS 1354-64 (=BW Cir) during hard outburst: evidence for optical cyclo-synchrotron emission from the hot accretion flow*
- Antia, H. M., Yadav, J. S., Agrawal, P. C., Verdhhan Chauhan, Jai, Manchanda, R. K., Chitnis, Varsha, Paul, Biswajit, Dedhia, Dhiraj, Shah, Parag, Gujar, V. M., Katoch, Tilak, Kurhade, V. N., Madhwani, Pankaj, Manojkumar, T. K., Nikam, V. A., Pandya, A. S., Parmar, J. V., Pawar, D., M., **Pahari, M.**, Misra, Ranjeev, et al., 2017, ApJS, 231, 10 : Link: <https://ui.adsabs.harvard.edu/abs/2017ApJS..231...10A> : *Calibration of the Large Area X-Ray Proportional Counter (LAXPC) Instrument on board AstroSat*
- Court, J. M. C., Altamirano, D., Pereyra, M., Boon, C. M., Yamaoka, K., Belloni, T., Wijnands, R., and **Pahari, M.** 2017, MNRAS, 468, 4748: Link: <https://ui.adsabs.harvard.edu/abs/2017MNRAS.468.4748C> : *An atlas of exotic variability in IGR J17091-3624: a comparison with GRS 1915+105*
- Verdhhan Chauhan, Jai, Yadav, J. S., Misra, Ranjeev, Agrawal, P. C., Antia, H. M., **Pahari, M.**, Sridhar, Navin, et al. 2017, ApJ, 841, 41: Link: <https://ui.adsabs.harvard.edu/abs/2017ApJ...841...41V> : *AstroSat/LAXPC Detection of Millisecond Phenomena in 4U*

1728-34

- Mallick, Labani, Dewangan, Gulab C., McHardy, I. M., & Pahari, M., 2017, MNRAS, 472, 174: Link: <https://ui.adsabs.harvard.edu/abs/2017MNRAS.472..174M> : *Energy-dependent variability of the bare Seyfert 1 galaxy Ark 120*
- Mondal, Aditya S., **Pahari, M.**, Dewangan, G. C., Misra, R., and Raychaudhuri, B. 2017, MNRAS, 466, 4991: Link: <https://ui.adsabs.harvard.edu/abs/2017MNRAS.466.4991M> : *NuSTAR and Swift joint view of neutron star X-ray binary 4U 1728-34: disc reflection in the island and lower banana states*
- Misra, Ranjeev, Yadav, J. S., Verdhhan Chauhan, Jai, Agrawal, P. C., Antia, H. M., **Pahari, M.**, Chitnis, V. R., Dedhia, Dhiraj, et al. 2017, ApJ, 835, 195 : Link: <https://ui.adsabs.harvard.edu/abs/2017ApJ...835..195M> : *AstroSat/LAXPC Observation of Cygnus X-1 in the Hard State*
- Gandhi, P., Littlefair, S. P., Hardy, L. K., Dhillon, V. S., Marsh, T. R., Shaw, A. W., Altamirano, D., Caballero-Garcia, M. D., Casares, J., Casella, P., Castro-Tirado, A. J., Charles, P. A., Dallilar, Y., Eikenberry, S., Fender, R. P., Hynes, R. I., Knigge, C., Kuulkers, E., Mooley, K., Munoz-Darias, T., **Pahari, M.**, Rahoui, F., et al. 2016, MNRAS, 459, 554: Link: <https://ui.adsabs.harvard.edu/abs/2016MNRAS.459..554G> : *Furiously fast and red: sub-second optical flaring in V404 Cyg during the 2015 outburst peak*
- Yadav, J. S., Misra, Ranjeev, Verdhhan Chauhan, Jai, Agrawal, P. C., Antia, H. M., **Pahari, M.**, Dedhia, Dhiraj, et al. 2016, ApJ, 833, 27: Link: <https://ui.adsabs.harvard.edu/abs/2016ApJ...833...27Y> : *Astrosat/LAXPC Reveals the High-energy Variability of GRS 1915+105 in the chi Class*
- Mondal, Aditya S., Dewangan, G. C., **Pahari, M.**, Misra, R., Kembhavi, A. K., Raychaudhuri, B. 2016, MNRAS, 461, 191: Link: <https://ui.adsabs.harvard.edu/abs/2016MNRAS.461.1917M> : *Broad-band X-ray emission and the reality of the broad iron line from the neutron star-white dwarf X-ray binary 4U 1820-30*
- Mir, Mubashir Hamid, Misra, Ranjeev, **Pahari, M.**, Iqbal, Naseer, Ahmad, Naveel 2016, MNRAS, 457, 2999: Link: <https://ui.adsabs.harvard.edu/abs/2016MNRAS.457.2999M> : *A model for the energy-dependent time-lag and rms of the heartbeat oscillations in GRS 1915+105*
- **Pahari, M.**, Misra, Ranjeev, Dewangan, Gulab C., and Pawar, Pramod 2015, ApJ, 814, 158: Link: <https://ui.adsabs.harvard.edu/abs/2015ApJ...814..158P> : *Constraining Distance and Inclination Angle of V4641 Sgr Using Swift and NuSTAR Observations during Low Soft Spectral State*
- **Pahari, M.**, Yadav, J. S., and Bhattacharyya, Sudip 2014, ApJ, 783, 141: Link: <https://ui.adsabs.harvard.edu/abs/2014ApJ...783..141P> : *X-Ray Spectral State Evolution in IGR J17091-3624 and Comparison of its Heartbeat Oscillation Properties with those of GRS 1915+105*
- **Pahari, M.**, Misra, Ranjeev, Mukherjee, Arunava, Yadav, J. S., and Pandey, S. K. 2013, MNRAS, 436, 2334: Link: <https://ui.adsabs.harvard.edu/abs/2013MNRAS.436.2334P> : *Interpreting the large amplitude X-ray variation of GRS 1915+105 and IGR J17091-3624*

as modulations of an accretion disc

- **Pahari, M.**, Neilsen, Joseph, Yadav, J. S., Misra, Ranjeev, & Uttley, Phil 2013, ApJ, 778, 136 Link: <https://ui.adsabs.harvard.edu/abs/2013ApJ...778..136P> : *Comparison of Time/Phase Lags in the Hard State and Plateau State of GRS 1915+105*
- **Pahari, M.**, Yadav, J. S., Rodriguez, J., Misra, Ranjeev, Bhattacharyya, Sudip, and Pandey, S. K. 2013, ApJ, 778, 46 : Link: <https://ui.adsabs.harvard.edu/abs/2013ApJ...778...46P> : *Properties of Unique Hard X-Ray Dips Observed from GRS 1915+105 and IGR J17091-3624 and Their Implications*
- **Pahari, M.** and Pal, S. 2012, MNRAS, 423, 3352 : Link: <https://ui.adsabs.harvard.edu/abs/2012MNRAS.423.3352P> : *RXTE observation of recent flaring activity from the transient X-ray pulsar 2S 1553-542*
- **Pahari, M.**, Bhattacharyya, Sudip, Yadav, J. S., AND Pandey, S. K. 2012, MNRAS Letters, 422, 87 : Link: <https://ui.adsabs.harvard.edu/abs/2012MNRAS.422L..87P> : *Evidence of two unique variability classes from IGR J17091-3624*
- **Pahari, M.** and Pal, Sabyasachi 2010, MNRAS, 409, 903: Link: <https://ui.adsabs.harvard.edu/abs/2010MNRAS.409..903P> : *Signature of long-term class evolution in GRS 1915+105 at a high accretion rate*
- **Pahari, M.**, Yadav, J. S., Mishra, Suprabha A., and Pandya, A., 2010, NIMPA, 621, 364: Link: <https://ui.adsabs.harvard.edu/abs/2010NIMPA.621..364P> : *Large area high temperature hard X-ray spectroscopy detectors for space experiments*

Selected non-refereed Publication

- Yadav, J. S., Agrawal, P. C., Antia, H. M., Chauhan, Jai Verdhana, Dedhia, D., K., Tilak, M., P., Manchanda, R. K., Misra, R., **Pahari, M.**, et al., 2016, Space Telescopes and Instrumentation 2016: Ultraviolet to Gamma Ray :Link: <https://ui.adsabs.harvard.edu/abs/2016SPIE.9905E..1DY> : *Large Area X-ray Proportional Counter (LAXPC) instrument onboard ASTROSAT*
- **Pahari, M.**, Singh Yadav, J., Antia, H. M., and Misra, R. 2016, 41st COSPAR Scientific Assembly : Link: <https://ui.adsabs.harvard.edu/abs/2016cosp...41E1495P> : *ASTROSAT/LAXPC observations of X-Ray binaries: A new window to hard X-Ray science aspects*
- Yadav, J. S. And **Pahari, M.** 2011, Astronomical Society of India Conference Series: Link: <https://ui.adsabs.harvard.edu/abs/2011ASInC...3..136Y> : *Stable X-ray states in X-ray black hole binaries*
- **Pahari, M.**, Bhattacharyya, S., and Yadav, J. S. 2011, The Astronomer's Telegram, 1 Link: <https://ui.adsabs.harvard.edu/abs/2011ATel.3667....1P> : *Black hole X-ray binary IGR J17091-3624 shows a unique intensity variation pattern*
- **Pahari, M.**, Pal, S. and Mukherjee, A. 2010, The Astronomer's Telegram, 1 : Link: <https://ui.adsabs.harvard.edu/abs/2010ATel.2902....1P> : *RXTE/PCA detection of QPO near 0.22 Hz in Aql X-1 during ongoing outburst*